

On relative Koszul duality and cluster categories

Li Fan

joint work with Bernhard Keller and Yu Qiu
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Morningside Center of Mathematics, Chinese Academy of Sciences

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Orbit cluster categories

Let $\mathcal{T}$ be a triangulated category and $F : \mathcal{T} \rightarrow \mathcal{T}$ be a triangulated equivalence functor. The orbit category $\mathcal{T}/F$ is the additive category whose

- objects are the same as the objects of $\mathcal{T}$, and
- morphisms are given by $\mathcal{T}/F(X, Y) = \bigoplus_{p \in \mathbb{Z}} \mathcal{T}(X, F^p Y)$.

An example: let A be a finite-dimensional hereditary $\mathbf{k}$ -algebra.

- Buan-Marsh-Reineke-Reiten-Todorov (2005) defined the cluster category as the orbit category

$$\mathcal{C}_2(A) := \mathcal{D}^b(\text{mod } A) / \mathbb{S} \circ [-2],$$

where $\mathbb{S} := - \bigotimes_A^L DA$ is the Serre functor and $DA = \text{Hom}_{\mathbf{k}}(A, \mathbf{k})$. We have the following relation

$$\mathbb{S} = \tau^{-1} \circ [1],$$

where τ is the AR-translation functor.

Triangulated hulls of orbit categories

- In general, the orbit categories of triangulated categories not necessarily carry a triangulated structure.
- For example, when hereditary assumption is removed (ex. $\text{gldim } A < \infty$), the orbit category $\text{per } A/\Sigma_m$ is no longer triangulated, where $\Sigma_m := \mathbb{S} \circ [-m]$ is the auto-equivalence of $\text{per } A$ and m is an integer ≥ 2 . However, it has a triangulated hull (Keller)

$$\mathcal{C}_m(A) := \langle A \rangle_B / \text{per } B,$$

where B is the dg algebra $A \oplus DA[-m-1]$. Moreover, when A is hereditary, $\mathcal{C}_m(A)$ coincides with $\text{per } A/\Sigma_m$.

Perfect and perfectly valued derived categories

Let A be a dg algebra over $\mathbf{k}$.

- The perfect derived category is the full subcategory of compact objects:

$$\text{per}(A) := \mathcal{D}(A)^c = \text{thick}_{\mathcal{D}(A)}(A).$$

- The perfectly valued derived category is

$$\text{pvd}(A) := \{M \in \mathcal{D}(A) \mid M \text{ is perfect as a dg } \mathbf{k}\text{-module}\}.$$

Since $\mathbf{k}$ is a field, this is equivalent to

$$\sum_{p \in \mathbb{Z}} \dim_{\mathbf{k}} H^p(M) < \infty.$$

- If $A = A^0$ is an ordinary finite-dimensional algebra, then

$$\text{per}(A) \simeq \mathcal{K}^b(\text{proj } A), \quad \text{pvd}(A) = \mathcal{D}^b(\text{mod } A).$$

cosingularity categories versus singularity categories

Let $A^e = A^{op} \otimes_k A$ be the enveloping algebra.

- A is connective if $H^i(A) = 0$ for all $i > 0$.
- A is smooth if A , viewed as an A - A -bimodule, is perfect in $\mathcal{D}(A^e)$:

$$A \in \text{per } A^e.$$

In this case, every perfectly valued dg A -module is perfect:

$$\text{pvd}(A) \subseteq \text{per}(A) \implies \text{cosg}(A) := \frac{\text{per}(A)}{\text{pvd}(A)} \quad (\text{cosingularity category})$$

- A is proper if A , viewed as a dg $\mathbf{k}$ -module, is perfect: $A \in \text{per } \mathbf{k}$.
Since $\text{pvd}(A)$ is thick and contains A , we have

$$\text{per}(A) \subseteq \text{pvd}(A) \implies \text{sg}(A) := \frac{\text{pvd}(A)}{\text{per}(A)} \quad (\text{singularity category}).$$

- If $A = A^0$ is finite-dimensional and self-injective, then

$$\text{sg}(A) \simeq \underline{\text{mod}}(A).$$

Generalized cluster categories

Amiot-Guo-Keller defined a generalized version of cluster category. Let A be a connective, smooth and proper dg algebra over $\mathbf{k}$, and $N \geq 3$.

- The N -Calabi-Yau completion of A is the derived dg tensor algebra

$$\Pi_N A = T_A(\Theta) = A \oplus \Theta \oplus (\Theta \otimes_A \Theta) \oplus \cdots,$$

where $\Theta = \theta[N-1]$, θ is a cofibrant replacement of the inverse dualizing complex $A^\vee := \mathrm{RHom}_{A^e}(A, A^e)$.

- $\mathcal{C}(\Pi_N A) := \mathrm{per}(\Pi_N A) / \mathrm{pvd}(\Pi_N A)$, the generalized $(N-1)$ -cluster category.
- $\mathrm{pvd}(\Pi_N A)$ is N -Calabi-Yau and if $\mathcal{C}(\Pi_N A)$ is Hom-finite, then there is a triangle equivalence

$$\mathcal{C}(\Pi_N A) \simeq \mathcal{C}_{N-1}(A).$$

Furthermore, $\mathcal{C}(\Pi_N A)$ is $(N-1)$ -Calabi-Yau.

Differential bigraded categories/algebras

- Let $\mathbf{k}$ be an algebraically closed field.
- Let $\text{Grm}(\mathbf{k})$ be the category of $\mathbb{Z}$ -graded $\mathbf{k}$ -modules $M = \bigoplus_{p \in \mathbb{Z}} M_p$ with grading shift given by $(M(1))_p = M_{p+1}$.
- Let $\mathcal{C}_{dbg}(\mathbf{k})$ be the dg category of complexes over $\text{Grm}(\mathbf{k})$.
- The differential bigraded (dbg) category $\mathcal{A}$ is a category enriched in $\mathcal{C}_{dbg}(\mathbf{k})$.

Remark

If $\mathcal{A}$ is a dbg category with one object $*$, then we can identify $\mathcal{A}$ with the dbg algebra $A = \mathcal{A}(*, *)$, i.e. a dg $\mathbf{k}$ -algebra endowed with the Adams $\mathbb{Z}$ -grading

$$A = \bigoplus_{p \in \mathbb{Z}} A_p$$

such that $|d| = (1, 0)$.

The derived categories

The bigraded derived category $\mathcal{D}^{\mathbb{Z}}(\mathcal{A})$ is the category of dbg modules $M : \mathcal{A}^{op} \rightarrow \mathcal{C}_{dbg}(\mathbf{k})$ localized at the $s : M \rightarrow L$ such that $s_p X : L_p X \rightarrow M_p X$ is a quasi-isom for any $p \in \mathbb{Z}$ and $X \in \text{obj}(\mathcal{A})$.

Remark

$\mathcal{D}^{\mathbb{Z}}(\mathcal{A})$ is compactly generated by $\{X^\wedge(p), p \in \mathbb{Z}, X \in \text{obj}(\mathcal{A})\}$, where $X^\wedge := \mathcal{A}(?, X)$.

The perfect derived category is

$$\text{per}^{\mathbb{Z}}(\mathcal{A}) := \text{thick}\{X^\wedge(p), p \in \mathbb{Z}, X \in \text{obj}(\mathcal{A})\}.$$

The perfectly valued derived category is

$$\text{pvd}^{\mathbb{Z}}(\mathcal{A}) := \{M \in \mathcal{D}^{\mathbb{Z}}(\mathcal{A}) \mid MX \in \text{per}^{\mathbb{Z}}(\mathbf{K}), \forall X \in \text{obj}(\mathcal{A})\},$$

where $\mathbf{K}$ is $\mathbf{k}$ when regarded as a dbg algebra concentrated in bidegree $(0, 0)$.

$\mathbb{X}$ -Calabi–Yau completion

Let A be a smooth, proper and connective dg algebra.

- A triangulated category $\mathcal{D}$ is called $\mathbb{X}$ -Calabi–Yau if its Serre functor is (1) , i.e.

$$\mathrm{Hom}_{\mathcal{D}}(L, M) \cong D \mathrm{Hom}_{\mathcal{D}}(M, L(1))$$

naturally for any $L, M \in \mathcal{D}$.

- The $\mathbb{X}$ -Calabi–Yau completion of A is the differential bigraded tensor algebra

$$\Pi_{\mathbb{X}} A = T_A(A^\vee[-1](1)) = \bigoplus_{p \geq 0} (A^\vee[-1](1))^{\otimes_A^p},$$

where $A^\vee$ is taken up to a cofibrant replacement.

- By smoothness of A , $\Pi_{\mathbb{X}} A$ is smooth, and $\mathrm{pvd}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A)$ is a full subcategory of $\mathrm{per}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A)$. Moreover, $\mathrm{pvd}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A)$ is an $\mathbb{X}$ -Calabi–Yau category.

An example: A_2 case $(1, -1) \begin{array}{c} \circlearrowleft \\ \circlearrowright \end{array} 1 \begin{array}{c} \xrightarrow{(0,0)} \\ \xleftarrow{(2,-1)} \end{array} 2 \begin{array}{c} \circlearrowleft \\ \circlearrowright \end{array} (1, -1)$

Ginzburg dbg algebra

Definition

Let $Q = (Q_0, Q_1)$ be a finite acyclic quiver. The Ginzburg Calabi-Yau- $\mathbb{X}$ dbg algebra $\Gamma_{\mathbb{X}}Q := (\mathbf{k}\overline{Q}, d)$ is defined as follows. The $\mathbb{Z}^2$ -graded quiver $\overline{Q}$ with vertices Q_0 and following arrows:

- original arrows $a : i \rightarrow j \in Q_1$ with bidegree $(0, 0)$;
- opposite arrows $a^* : j \rightarrow i \in Q_1$ with bidegree $(2, -1)$;
- a loop e_i^* for each $i \in Q_0$ with bidegree $(1, -1)$.

Define a differential $d : \mathbf{k}\overline{Q} \rightarrow \mathbf{k}\overline{Q}$ of bidegree $(1, 0)$ by

- $da = da^* = 0$ for $a \in Q_1$;
- $de_i^* = e_i \left(\sum_{a \in Q_1} (aa^* - a^*a) \right) e_i$.

Notice that in this case, we have $\Pi_{\mathbb{X}}(\mathbf{k}Q) \cong \Gamma_{\mathbb{X}}Q$.

Perfect derived category as ∞ -cluster category

Theorem (Ikeda-Qiu)

We have a triangle equivalence

$$\begin{array}{ccc}
 \mathbb{S} = ? \overset{L}{\otimes}_A DA & \xrightarrow{\quad} & [-1](1) \\
 \downarrow \text{per } A & \sim & \downarrow \\
 \text{per } A & \xrightarrow{\quad} & \mathcal{C}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A).
 \end{array} \tag{1}$$

Note that $\mathbb{S} = \tau^{-1} \circ [1]$ is the Serre functor.

Perfect derived category as singularity categories

- Let A be a f.d. algebra with $\text{gldim } A < \infty$.
- Let $E_{\mathbb{X}} = A \oplus DA(-1)$ be the trivial extension of A .
- $\text{mod}^{\mathbb{Z}}(E_{\mathbb{X}})$ is equivalent to $\text{mod}(\hat{A})$, where $\hat{A}$ is the repetitive algebra.

Theorem (Happel, Rickard)

There is an equivalence

$$\begin{array}{ccc}
 \mathbb{S} \circ [1] & \xrightarrow{\quad} & (1) \\
 \downarrow \text{hook} & & \downarrow \text{hook} \\
 \text{per } A & \xrightarrow{\quad} & \underline{\text{mod}}^{\mathbb{Z}}(E_{\mathbb{X}}) \rightarrow \text{sg}^{\mathbb{Z}}(E_{\mathbb{X}}) := \text{pvd}^{\mathbb{Z}}(E_{\mathbb{X}}) / \text{per}^{\mathbb{Z}}(E_{\mathbb{X}}).
 \end{array}
 \quad \sim \quad$$

(2)

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Two differential bigraded algebras

Let A be a dg algebra and write $A^e = A \otimes_k A^{op}$. Let $X \in \mathcal{D}(A^e)$ an invertible dg bimodule with inverse Y , i.e.

$$X \overset{L}{\otimes}_A Y \simeq A \text{ and } Y \overset{L}{\otimes}_A X \simeq A$$

in $\mathcal{D}^{\mathbb{Z}}(A^e)$. Write $\hat{X}$ for a cofibrant resolution of $X[-1](1)$ and let

$$T = T_A(\hat{X}) = \bigoplus_{p \geq 0} \hat{X}^{\otimes_A^p} \quad (3)$$

be the differential bigraded tensor algebra of $\hat{X}$ over A and

$$E = A \oplus Y(-1) \quad (4)$$

the differential bigraded trivial extension algebra of $Y(-1)$ by A .

The relative perfectly valued derived category

Let A be a dg algebra, B a dbg algebra and $i_A^B : A \rightarrow B$ a morphism of dbg algebras. There is an induced restriction functor:

$$\mathcal{D}^{\mathbb{Z}}(B) \rightarrow \mathcal{D}^{\mathbb{Z}}(A).$$

Definition

We define the relative perfectly valued derived categories of B , with respect to A , to be

$$\mathrm{pvd}^{\mathbb{Z}}(B, A) := \{M \in \mathcal{D}^{\mathbb{Z}}(B) \mid M|_A \in \mathrm{per}^{\mathbb{Z}}(A)\}. \quad (5)$$

Notice that $\mathrm{pvd}^{\mathbb{Z}}(T, A)$ and $\mathrm{pvd}^{\mathbb{Z}}(E, A)$ equal the thick subcategory of $\mathcal{D}^{\mathbb{Z}}(T)$ and $\mathcal{D}^{\mathbb{Z}}(E)$ generated by the $A(q)$, $q \in \mathbb{Z}$ respectively.

Enlarged cluster categories

Definition

We define the enlarged cluster category of A wrt X as the Verdier quotient

$$\mathcal{C}^{\mathbb{Z}}(T, A) := \text{per}^{\mathbb{Z}}(T) / \text{pvd}^{\mathbb{Z}}(T, A). \quad (6)$$

A generalization of a theorem by Ikeda-Qiu:

Theorem (Ikeda-Qiu, F-Keller-Qiu)

The composition

$$\begin{array}{ccc} ? \overset{L}{\otimes}_A Y[1] & \xrightarrow{\quad} & (1) \\ \downarrow & & \downarrow \\ \Phi : \text{per } A & \xrightarrow{? \overset{L}{\otimes}_A T} \text{per}^{\mathbb{Z}}(T) & \longrightarrow \mathcal{C}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) \end{array}$$

is a triangle equivalence. Moreover, we have $(1) \circ \Phi \simeq \Phi \circ (? \overset{L}{\otimes}_A Y[1])$.

Shrunk singularity categories

Definition

We define the shrunk singularity category of A as the Verdier quotient

$$\mathrm{sg}^{\mathbb{Z}}(E, A) := \mathrm{pvd}^{\mathbb{Z}}(E, A) / \mathrm{per}^{\mathbb{Z}}(E). \quad (7)$$

A generalization of Happel's description of the derived category:

Theorem (Hanihara, F-Keller-Qiu)

The restriction along the augmentation $E \rightarrow A$ induces an equivalence

$$\begin{array}{ccc} ? \overset{L}{\otimes}_A Y[1] & \xrightarrow{\quad} & (1) \\ \downarrow \scriptstyle \text{hook} & & \downarrow \scriptstyle \text{hook} \\ \mathrm{per} A & \xrightarrow{\quad \sim \quad} & \mathrm{sg}^{\mathbb{Z}}(E, A). \end{array} \quad (8)$$

Moreover, we have $(1) \circ \Psi \simeq \Psi \circ (? \overset{L}{\otimes}_A Y[1])$.

Relative Koszul duality

Theorem (F-Keller-Qiu)

The adjoint pair

$$\text{per}^{\mathbb{Z}}(T) \begin{array}{c} \xrightarrow{\text{RHom}_T^{\mathbb{Z}}(A, ?)} \\ \xleftarrow{? \otimes_E^L A} \end{array} \text{pvd}^{\mathbb{Z}}(E, A). \quad (9)$$

induces the following commutative diagram

$$\begin{array}{ccccc} \text{pvd}^{\mathbb{Z}}(T, A) & \hookrightarrow & \text{per}^{\mathbb{Z}}(T) & \longrightarrow & \mathcal{C}^{\mathbb{Z}}(T, A) \\ \downarrow \wr & & \downarrow \wr \text{RHom}_T^{\mathbb{Z}}(A, ?) & & \downarrow \wr \\ \text{per}^{\mathbb{Z}}(E) & \hookrightarrow & \text{pvd}^{\mathbb{Z}}(E, A) & \longrightarrow & \text{sg}^{\mathbb{Z}}(E, A) \end{array} \quad \begin{array}{l} \swarrow [1] \circ \Phi, \sim \\ \searrow \Psi, \sim \end{array} \text{per } A, \quad (10)$$

where "all" functors commute with (1).

Example: Classical Koszul duality

For the case when $A = \mathbf{k}$, $X = \mathbf{k}[1](-1)$ and $Y = \mathbf{k}[-1](1)$, we have

$$T = T_A(\mathbf{k}) = \mathbf{k}[u],$$

where u is of bidegree $(0, 0)$ and

$$E = \mathbf{k} \oplus \mathbf{k}[-1] = \mathbf{k}[v]/(v^2),$$

where v is of bidegree $(0, 0)$. Here we consider T and E as augmented dg A -algebras with vanishing differential. We see that T and E are Koszul dual to each other in the sense that

$$\mathrm{RHom}_E(A, A) \simeq T$$

and

$$\mathrm{RHom}_T(A, A) \simeq E.$$

Example: The Calabi-Yau-X case

If A is a smooth, proper and connective dg algebra, then we can take

$$X = A^\vee \text{ and } Y = DA.$$

In this case, $T = \Pi_{\mathbb{X}} A$ is the $\mathbb{X}$ -Calabi-Yau completion, and $E_{\mathbb{X}} = A \oplus DA(-1)$ is the trivial extension of A respectively. Moreover, the enlarged cluster category reduces to the ∞ -cluster category

$$\mathcal{C}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) := \text{per}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) / \text{pvd}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A)$$

and the shrunk singularity category becomes

$$\text{sg}^{\mathbb{Z}}(E_{\mathbb{X}}) := \text{pvd}^{\mathbb{Z}}(E_{\mathbb{X}}) / \text{per}^{\mathbb{Z}}(E_{\mathbb{X}}).$$

The theorem generalizes Ikeda-Qiu's and Happel's results:

$$\text{per } A \xrightarrow{\sim} \mathcal{C}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) \text{ and } \text{per } A \xrightarrow{\sim} \text{sg}^{\mathbb{Z}}(E_{\mathbb{X}}),$$

which realize the perfect derived category as ∞ -cluster category and singularity category respectively and build a connection between them.

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Step 1: the left lax quotient

Let $\mathcal{A}$ be a dg category and $F \in \text{rep}_{dg}(\mathcal{A}, \mathcal{A})$ a dg bimodule.

The left lax quotient $\mathcal{A}/_{//} F^{\mathbb{N}}$ of $\mathcal{A}$ by F is the dg category whose

- objects are the same as the objects of $\mathcal{A}$, and
- morphisms are given by $\mathcal{A}/_{//} F^{\mathbb{N}}(X, Y) = \bigoplus_{p \in \mathbb{N}} \mathcal{A}(X, F^p Y)$.
- Composition is induced by

$$\begin{array}{l} f : X \rightarrow F^p Y, \\ g : Y \rightarrow F^q Z \end{array} \rightsquigarrow F^p(g) \circ f : X \rightarrow F^{p+q} Z.$$

The canonical dg functor $Q_{\mathbb{N}} : \mathcal{A} \rightarrow \mathcal{A}/_{//} F^{\mathbb{N}}$ acts on

- objects: it sends $X \in \text{obj}(\mathcal{A})$ to $X \in \text{obj}(\mathcal{A}/_{//} F^{\mathbb{N}})$, and
- morphisms: it sends $f : X \rightarrow Y$ to $f \in \mathcal{A}(X, Y) \subseteq \bigoplus_{p \in \mathbb{N}} \mathcal{A}(X, F^p Y)$.

The canonical morphism of dg functors $q : Q_{\mathbb{N}} F \rightarrow Q_{\mathbb{N}}$ acts on the objects of $\mathcal{A}$ as

$$qX := \text{id}_{FX} \in \mathcal{A}(FX, FX) \subseteq \bigoplus_{p \in \mathbb{N}} \mathcal{A}(FX, F^{p+1} X).$$

Step 2: localize to get the dg orbit category

Definition

The dg orbit category $\mathcal{A}/F^{\mathbb{Z}}$ is the dg localization

$$\mathcal{A}/F^{\mathbb{Z}} = (\mathcal{A}/_{//} F^{\mathbb{N}})[q^{-1}]$$

of $\mathcal{A}/_{//} F^{\mathbb{N}}$ with respect to the morphisms

$$qX : Q_{\mathbb{N}}FX \rightarrow Q_{\mathbb{N}}X, \quad X \in \text{obj}(\mathcal{A}).$$

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{Q_{\mathbb{N}}} & \mathcal{A}/_{//} F^{\mathbb{N}} \\ & \searrow Q_{\mathbb{Z}} & \downarrow Q_{\text{loc}} \\ & & \mathcal{A}/F^{\mathbb{Z}} \end{array}$$

When F induces an equivalence $H^0 F : H^0 \mathcal{A} \rightarrow H^0 \mathcal{A}$

- For $X, Y \in \text{obj}(\mathcal{A})$, the morphism space $\mathcal{A}/F^{\mathbb{Z}}(X, Y)$ is quasi-isomorphic to the direct sum of the complexes $\mathcal{A}(X, F^p Y)$ for all $p \geq 0$ and $\mathcal{A}(F^{-p} X, Y)$ for all $p < 0$.
- We have an isomorphism

$$H^0(\mathcal{A}/F^{\mathbb{Z}})(X, Y) = \bigoplus_{p \in \mathbb{Z}} (H^0 \mathcal{A})(X, (H^0 F)^p Y).$$

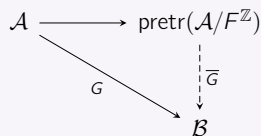
Universal property: equivariant functors

$\mathbb{Z}$ -equivariant functors

An object of $\mathbb{Z}\text{-Eq}(\mathcal{A}, F, \mathcal{B})$ is a pair (G, γ) where

$$G \in \text{rep}_{dg}(\mathcal{A}, \mathcal{B}), \quad \gamma : GF \longrightarrow G,$$

and each γ_X is an isomorphism in $H^0(\mathcal{B})$.



Theorem (F-Keller-Qiu)

Let $\mathcal{A}, \mathcal{B}$ be two pretriangulated dg categories. Then $\mathcal{A} \rightarrow \text{pretr}(\mathcal{A}/F^{\mathbb{Z}})$ induces an isomorphism

$$\text{rep}_{dg}(\text{pretr}(\mathcal{A}/F^{\mathbb{Z}}), \mathcal{B}) \xrightarrow{\sim} \mathbb{Z}\text{-Eq}(\mathcal{A}, F, \mathcal{B}). \quad (11)$$

in $\text{Hqe} = \text{Ho}(\text{dgcats})$.

Triangulated orbit categories

Definition

Given a triangulated category $\mathcal{T}$ with a dg enhancement $H^0(\mathcal{A}) \xrightarrow{\sim} \mathcal{T}$ and a dg bimodule $F \in \text{rep}_{dg}(\mathcal{A}, \mathcal{A})$ inducing an equivalence $H^0(F) : H^0(\mathcal{A}) \rightarrow H^0(\mathcal{A})$, then the triangulated orbit category of $\mathcal{T}$ wrt $\mathcal{A}$ is defined as

$$H^0(\text{pretr}(\mathcal{A}/F^{\mathbb{Z}})).$$

Example: Let A be a connective, smooth and proper dg algebra and $m \geq 2$ be an integer.

- $\text{per } A$ has a canonical dg enhancement given by the dg category $\mathcal{A}$ of perfect cofibrant dg A -modules. We take $X \rightarrow DA$, a cofibrant resolution of DA in the category of dg A -bimodules.

Construction of the orbit m -cluster category

We take $\mathfrak{S}_m := - \otimes_A X[-m]$ to be the tensor product

$$\begin{aligned} \mathfrak{S}_m: \mathcal{A} &\rightarrow \mathcal{A} \\ P &\mapsto P \otimes_A X[-m]. \end{aligned} \tag{12}$$

We notice that $\mathfrak{S}_m$ itself is not a dg equivalence but the induced triangle functor $H^0(\mathfrak{S}_m)$ is an equivalence.

Definition

The orbit m -cluster category associated to A is defined as the triangulated orbit category $H^0(\text{pretr}(\mathcal{A}/\mathfrak{S}_m^{\mathbb{Z}}))$.

As shown in Keller's work, when A is hereditary, the dg orbit category $\mathcal{A}/\mathfrak{S}_m^{\mathbb{Z}}$ is already pretriangulated. So we have

$$H^0(\text{pretr}(\mathcal{A}/\mathfrak{S}_m^{\mathbb{Z}})) = H^0(\mathcal{A}/\mathfrak{S}_m^{\mathbb{Z}}) \simeq H^0(\mathcal{A})/H^0(\mathfrak{S}_m) \simeq \text{per } A/\Sigma_m. \tag{13}$$

Commutation with the dg quotient

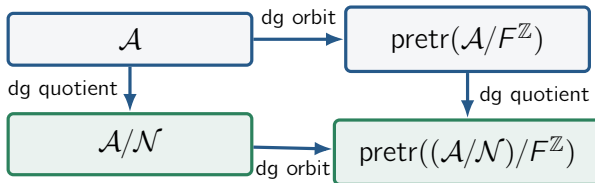
Suppose $\mathcal{N} \subseteq \mathcal{A}$ is a full dg subcategory such that $H^0(\mathcal{N})$ is stable under $H^0(F)$, so that F induces dg bimodules $F_{\mathcal{N}} \in \text{rep}_{dg}(\mathcal{N}, \mathcal{N})$ and $F_{\mathcal{A}/\mathcal{N}} \in \text{rep}_{dg}(\mathcal{A}/\mathcal{N}, \mathcal{A}/\mathcal{N})$. We have a canonical isomorphism in Hqe

$$\text{pretr}((\mathcal{A}/\mathcal{N})/F_{\mathcal{A}/\mathcal{N}}^{\mathbb{Z}}) \simeq \text{pretr}(\mathcal{A}/F^{\mathbb{Z}})/\text{pretr}(\mathcal{N}/F_{\mathcal{N}}^{\mathbb{Z}}). \quad (14)$$

Moreover, we have a short exact sequence of triangulated categories

$$0 \rightarrow (\mathcal{N}/F_{\mathcal{N}}^{\mathbb{Z}})^{tr} \rightarrow (\mathcal{A}/F^{\mathbb{Z}})^{tr} \rightarrow ((\mathcal{A}/\mathcal{N})/F_{\mathcal{A}/\mathcal{N}}^{\mathbb{Z}})^{tr} \rightarrow 0, \quad (15)$$

where $(-)^{tr}$ denotes the functor $H^0 \circ \text{pretr}$.



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Applications

Let $N \geq 3$ be an integer. There is a projection $\pi_N: \Pi_{\mathbb{X}} A \rightarrow \Pi_N A$ collapsing the bidegree $(a, b) \in \mathbb{Z}^2$ into $a + bN \in \mathbb{Z}$, that induces a functor $\pi_N: \text{per}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) \rightarrow \text{per}(\Pi_N A)$, which restricts to $\pi_N: \text{pvd}^{\mathbb{Z}} \Pi_{\mathbb{X}} A \rightarrow \text{pvd}(\Pi_N A)$.

Theorem (Ikeda-Qiu, F-Keller-Qiu)

Let A be a connective, smooth and proper dg algebra. We have the following commutative diagram between short exact sequences of triangulated categories:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \text{pvd}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) & \longrightarrow & \text{per}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) & \longrightarrow & \text{per } A \longrightarrow 0 \\
 & & \downarrow \mathbb{I}([-N](1)) & & \downarrow \mathbb{I}([-N](1)) & & \downarrow \mathbb{I}\mathbb{S}o[1-N] \\
 0 & \longrightarrow & \text{pvd}(\Pi_N A) & \longrightarrow & \text{per}(\Pi_N A) & \longrightarrow & \mathcal{C}_{N-1}(A) \longrightarrow 0.
 \end{array} \tag{16}$$

Dually, we also have a projection $\pi_N: E_{\mathbb{X}} \rightarrow E_N$ that induces functors between corresponding pvd and per, where $E_{\mathbb{X}} = A \oplus DA(-1)$ and $E_N = A \oplus DA[-N]$.

Combining everything

$$\begin{array}{ccccccc}
 \text{pvd}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) & \hookrightarrow & \text{per}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) & \longrightarrow & \mathcal{C}^{\mathbb{Z}}(\Pi_{\mathbb{X}} A) & & \\
 \swarrow \sim & \downarrow \text{//}([-M](1)) & \swarrow \sim & \downarrow \text{//}([-M](1)) & \swarrow \sim & \nwarrow \sim & \\
 \text{per}^{\mathbb{Z}}(E_{\mathbb{X}}) & \hookrightarrow & \text{pvd}^{\mathbb{Z}}(E_{\mathbb{X}}) & \longrightarrow & \text{sg}^{\mathbb{Z}}(E_{\mathbb{X}}) & \xleftarrow{\sim} & \text{per } A \\
 \downarrow \text{//}([-M](1)) & & \downarrow \text{//}([-M](1)) & & \downarrow \text{//}([-M](1)) & \downarrow \text{//}([-M](1)) & \downarrow \text{//}\mathbb{S} \circ [1-M] \\
 \text{pvd } \Pi_N A & \hookrightarrow & \text{per}(\Pi_N A) & \longrightarrow & \mathcal{C}(\Pi_N A) & & \\
 \swarrow \sim & \downarrow \text{//}([-M](1)) & \swarrow \sim & \downarrow \text{//}([-M](1)) & \swarrow \sim & \nwarrow \sim & \\
 \text{per}(E_N) & \hookrightarrow & \text{pvd}(E_N) & \longrightarrow & \text{sg}(E_N) & \xleftarrow{\sim} & \mathcal{C}_{N-1}(A)
 \end{array}$$

(17)

End

Thank you!

Main reference I

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Thanks!